

1.5.1. Let a card be selected from an ordinary deck of playing cards. The outcome c is one of these 52 cards. Let $X(c) = 4$ if c is an ace, let $X(c) = 3$ if c is a king, let $X(c) = 2$ if c is a queen, let $X(c) = 1$ if c is a jack, and let $X(c) = 0$ otherwise. Suppose that P assigns a probability of $\frac{1}{52}$ to each outcome c . Describe the induced probability $P_X(D)$ on the space $D = \{0, 1, 2, 3, 4\}$ of the random variable X .

$$P(X) = \frac{1}{52} \text{ (for any card)}$$

For $P_X(D)$ with $D=0$

$\Rightarrow$ cards that are not B, Q, K, A

$$\Rightarrow 52 - 4 \cdot 4 = 36$$

$$\Rightarrow P_X(D_0) = \frac{36}{52}$$

$$\left. \begin{array}{l} P_X(D_1) = \frac{4}{52} \\ \vdots \\ P_X(D_4) = \frac{4}{52} \end{array} \right\} \frac{16}{52}$$

$$P_X(D) = \frac{36}{52} + \frac{16}{52} = 1 //$$

$D_0 \dots D_4$

1.5.2. For each of the following, find the constant c so that $p(x)$ satisfies the condition of being a pmf of one random variable X .

(a) $p(x) = c(\frac{2}{3})^x$, $x = 1, 2, 3, \dots$, zero elsewhere.

(b) $p(x) = cx$, $x = 1, 2, 3, 4, 5, 6$, zero elsewhere.

$$a) c \cdot \left(\frac{2}{3} + \left(\frac{2}{3}\right)^2 + \dots \right) \stackrel{!}{=} 1$$

$$\Rightarrow \frac{2}{3}c \left(\underbrace{1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \dots}_{\text{geom. Series}} \right) = 1$$

$$\Rightarrow \frac{2}{3}c \left(\frac{1}{1 - \frac{2}{3}} \right) = 1$$

$$\Rightarrow \frac{2}{\cancel{3}}c \cdot \cancel{3} = 1$$

$$\Rightarrow 2c = 1$$

$$\Rightarrow c = \frac{1}{2} //$$

$$b) p(x) = cx, \quad x \in \{1, \dots, 6\}$$

$$\sum_1^6 cx \stackrel{!}{=} 1$$

$$\Rightarrow C \cdot \left[\frac{1}{2} x^2 \right]_1^6 = 1$$

$$\Rightarrow C \cdot \left(18 - \frac{1}{2} \right) = 1$$

$$\Rightarrow \frac{35}{2} C = 1$$

$$\Rightarrow C = \frac{2}{35} //$$

1.5.3. Let $p_X(x) = x/15$, $x = 1, 2, 3, 4, 5$, zero elsewhere, be the pmf of X . Find $P(X = 1 \text{ or } 2)$, $P(\frac{1}{2} < X < \frac{5}{2})$, and $P(1 \leq X \leq 2)$.

$$\begin{aligned} \Rightarrow P(X=1 \vee 2) &= \underbrace{P(X=1)}_{\frac{1}{15}} + \underbrace{P(X=2)}_{\frac{2}{15}} \\ &= \frac{3}{15} = \frac{1}{5} // \end{aligned}$$

$$\Rightarrow P\left(\frac{1}{2} < X < \frac{5}{2}\right) \rightarrow 0,5 < X < 2,5 \text{ gilt für } X=1, 2$$

$$\Rightarrow P(X=1) + P(X=2) = \frac{1}{5} //$$

AND

$$P(1 \leq X \leq 2) \rightarrow 1 \leq X \leq 2 \text{ gilt für } X=1,2$$

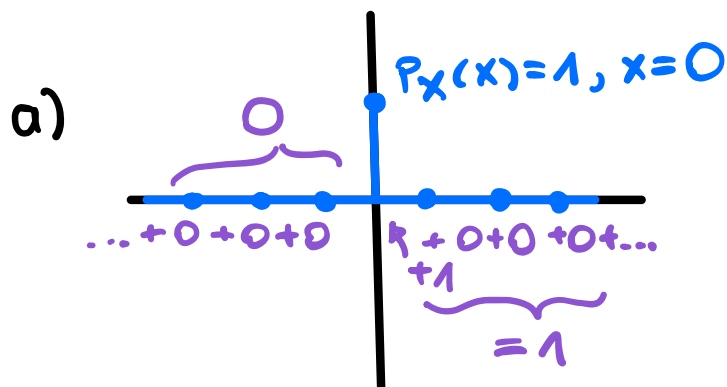
$$\Rightarrow P(X=1) + P(X=2) = \frac{1}{5}$$

1.5.4. Let $p_X(x)$ be the pmf of a random variable X . Find the cdf $F(x)$ of X and sketch its graph along with that of $p_X(x)$ if:

(a) $p_X(x) = 1, x = 0$, zero elsewhere.

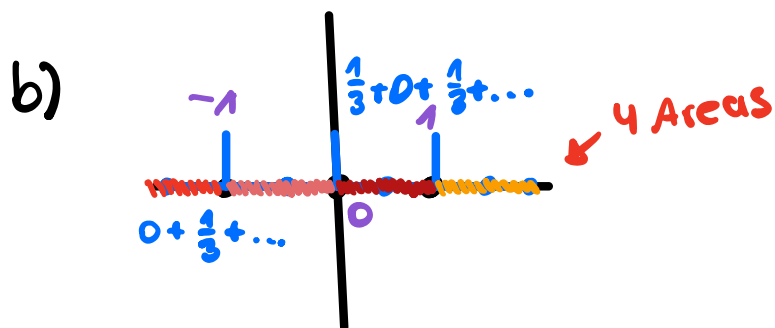
(b) $p_X(x) = \frac{1}{3}, x = -1, 0, 1$, zero elsewhere.

(c) $p_X(x) = x/15, x = 1, 2, 3, 4, 5$, zero elsewhere.



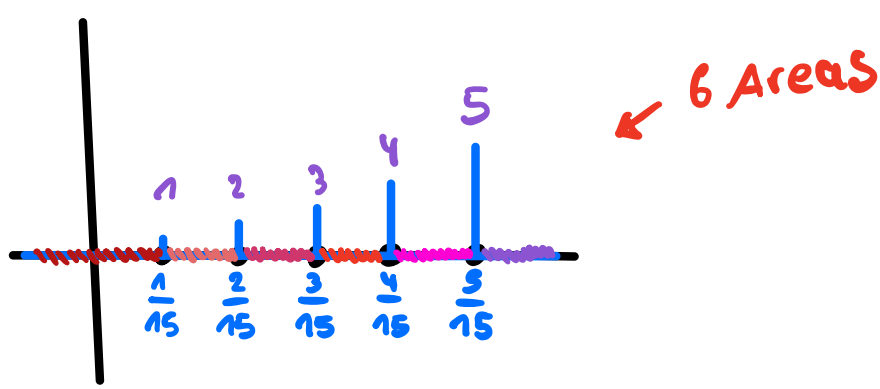
cumulative

$$\Rightarrow F(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$



$$\Rightarrow F(x) = \begin{cases} 0, & x < -1 \\ 1/3, & -1 \leq x < 0 \\ 2/3, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

c)



$$F(x) = \begin{cases} 0, & x < 1 \\ 1/15, & 1 \leq x < 2 \\ 3/15, & 2 \leq x < 3 \\ 6/15, & 3 \leq x < 4 \\ 10/15, & 4 \leq x < 5 \\ 1, & x \geq 5 \end{cases}$$

1.5.5. Let us select five cards at random and without replacement from an ordinary deck of playing cards.

(a) Find the pmf of X , the number of hearts in the five cards.

(b) Determine $P(X \leq 1)$.

a)
 $\Rightarrow$ # Ways to choose 5 cards : $\binom{52}{5}$

Choosing x ♥ : $\binom{13}{x}$

If we want 2 ♥ then we choose 3 remaining non-♥ cards.
 $52 - 13 = 39$ left

$\Rightarrow \binom{39}{5-x}$

This gives: $\frac{\binom{13}{x} \binom{39}{5-x}}{\binom{52}{5}}, x \in \{0, \dots, 5\}$

b) $P(X \leq 1) = P(X=0) + P(X=1)$

$$= \frac{\binom{13}{0} \binom{39}{5}}{\binom{52}{5}} + \frac{\binom{13}{1} \binom{39}{4}}{\binom{52}{5}}$$

$$= \frac{575757}{2598960} + \frac{1069263}{2598960}$$

$$\approx 63,3\%$$

1.5.6. Let the probability set function of the random variable X be $P_X(D) = \int_D f(x) dx$, where $f(x) = 2x/9$, for $x \in D = \{x : 0 < x < 3\}$. Define the events $D_1 = \{x : 0 < x < 1\}$ and $D_2 = \{x : 2 < x < 3\}$. Compute $P_X(D_1)$, $P_X(D_2)$, and $P_X(D_1 \cup D_2)$.

$$\bullet P_X(D) = \int_D f(x) dx$$

$$\bullet \int \frac{2}{9} x dx = \frac{1}{9} x^2$$

$$D_1 : P_X(D_1) = \int_0^1 \frac{2}{9} x^2 dx = \left[\frac{x^2}{9} \right]_0^1$$

$$= \frac{1}{9} - 0 = \frac{1}{9} //$$

$$\mathcal{D}_2 : P_X(\mathcal{D}_2) = \int_2^3 \frac{2}{9} x^2 dx = \left[\frac{x^3}{9} \right]_2^3$$

$$= \frac{9}{9} - \frac{4}{9} = \frac{5}{9} //$$

$x \in (0,1)$ $x \in (2,3)$ DISJOINT

$$\mathcal{D}_3 : P_X(\mathcal{D}_1 \cup \mathcal{D}_2) = P_X(\mathcal{D}_1) + P_X(\mathcal{D}_2)$$

$$= \frac{1}{9} + \frac{5}{9} = \frac{6}{9} = \frac{2}{3} //$$

1.5.7. Let the space of the random variable X be $\mathcal{D} = \{x : 0 < x < 1\}$. If $D_1 = \{x : 0 < x < \frac{1}{2}\}$ and $D_2 = \{x : \frac{1}{2} \leq x < 1\}$, find $P_X(D_2)$ if $P_X(D_1) = \frac{1}{4}$.

$$\rightarrow \mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset \wedge \mathcal{D}_1 \cup \mathcal{D}_2 = \mathcal{D}, P_X(\mathcal{D}) = 1$$

$$\Rightarrow \underbrace{P_X(\mathcal{D}_1)}_{\frac{1}{4}} + P_X(\mathcal{D}_2) = 1$$

$$\Rightarrow 1 - \frac{1}{4} = P_X(\mathcal{D}_2) = \frac{3}{4} //$$

1.5.8. Suppose the random variable X has the cdf

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x+2}{4} & -1 \leq x < 1 \\ 1 & 1 \leq x. \end{cases}$$

Write an R function to sketch the graph of $F(x)$. Use your graph to obtain the probabilities: (a) $P(-\frac{1}{2} < X \leq \frac{1}{2})$; (b) $P(X = 0)$; (c) $P(X = 1)$; (d) $P(2 < X \leq 3)$.

```
# A function to define and sketch the CDF
sketch_my_cdf <- function() {

  # 1. Define the piecewise CDF using Vectorize
  F_x <- Vectorize(function(x) {
    if (x < -1) {
      return(0)
    } else if (x < 1) {
      return((x + 2) / 4)
    } else {
      return(1)
    }
  })

  # 2. Plot the function
  curve(F_x, from = -3, to = 3, n = 1001,
        main = "Graph of the CDF F(x)",
        xlab = "x",
        ylab = "F(x) = P(X <= x)",
        lwd = 2, col = "blue")

  # 3. Add points to clarify the discontinuities
  # At x = -1, it jumps from 0 to 1/4
  points(-1, 0, pch = 1, cex = 1.5, col = "blue") # Open circle at
  (-1, 0)
  points(-1, 1/4, pch = 19, cex = 1.5, col = "blue") # Closed circle at
  (-1, 1/4)

  # At x = 1, it jumps from 3/4 to 1
  points(1, 3/4, pch = 1, cex = 1.5, col = "blue") # Open circle at ( 1,
  3/4 )
  points(1, 1, pch = 19, cex = 1.5, col = "blue") # Closed circle at
  ( 1, 1 )

  # Add gridlines for readability
  grid()
}

# --- To run this code, you would simply call the function: ---
# sketch_my_cdf()
```

$$a) F\left(\frac{1}{2}\right) = \frac{5/2}{4} = \frac{5}{8} \quad (b)$$

$$F\left(-\frac{1}{2}\right) = \frac{3/2}{4} = \frac{3}{8} \quad (a)$$

$$\Rightarrow F(b) - F(a) \rightarrow \frac{5}{8} - \frac{3}{8} = \frac{1}{4} //$$

$$b) F(0) = \frac{0+2}{4} = \frac{1}{2} \text{ and}$$

$$\lim_{x \rightarrow 0^-} \bar{F}(x) = \lim_{x \rightarrow 0^+} F(x) = \frac{1}{2} //$$

$$\left(\frac{0+2}{4}\right)$$

$$c) \bar{F}(1) = 1$$

$$\lim_{x \rightarrow 1^-} F(x) = \frac{1+2}{4} = \frac{3}{4} //$$

$$d) P(2 < X \leq 3)$$

$$\leadsto F(3) = 1 \\ F(2) = 1 \Rightarrow 1 - 1 = 0 //$$

1.5.9. Consider an urn that contains slips of paper each with one of the numbers $1, 2, \dots, 100$ on it. Suppose there are i slips with the number i on it for $i = 1, 2, \dots, 100$. For example, there are 25 slips of paper with the number 25. Assume that the slips are identical except for the numbers. Suppose one slip is drawn at random. Let X be the number on the slip.

(a) Show that X has the pmf $p(x) = x/5050$, $x = 1, 2, 3, \dots, 100$, zero elsewhere.

(b) Compute $P(X \leq 50)$.

(c) Show that the cdf of X is $F(x) = [x]([x] + 1)/10100$, for $1 \leq x \leq 100$, where $[x]$ is the greatest integer in x .

$$a) \# \text{ slips} = 1 + 2 + \dots + 100 \\ = \sum_{i=1}^{100} i = 5050$$

$$P(X=x) = \frac{\# \text{ slips } x}{\# \text{ total slips}} = \frac{x}{5050} //$$

$$b) P(X \leq 50)$$

$$\leadsto P(X=1) + P(X=2) + \dots + P(X=50)$$

$$= \frac{1}{5050} + \frac{2}{5050} + \dots + \frac{50}{5050}$$

$$= \sum_{k=1}^{50} \frac{k}{5050} = \frac{1}{5050} \sum_{k=1}^{50} k = \frac{1275}{5050} \approx 25.25\% //$$

$$c) F(x) = P(X \leq \lfloor x \rfloor) = P(X \leq k), \quad k = \lfloor x \rfloor$$

$$F(x) = P(X \leq x) := \text{CDF}$$

$$= \sum_{i=1}^k P(i) = \sum_{i=1}^k \frac{i}{5050}$$

$$= \frac{1}{5050} \sum_{i=1}^k i = \frac{1}{5050} \left(\frac{k(k+1)}{2} \right)$$

$$\stackrel{\text{resub.}}{=} \frac{1}{5050} \cdot \frac{(\lfloor x \rfloor (\lfloor x \rfloor + 1))}{2}$$

$$= \frac{\lfloor x \rfloor (\lfloor x \rfloor + 1)}{10100} //$$

1.5.10. Prove parts (b) and (c) of Theorem 1.5.1.

Theorem 1.5.1. Let X be a random variable with cumulative distribution function $F(x)$. Then

(a) For all a and b , if $a < b$, then $F(a) \leq F(b)$ (F is nondecreasing).

(b) $\lim_{x \rightarrow -\infty} F(x) = 0$ (the lower limit of F is 0).

(c) $\lim_{x \rightarrow \infty} F(x) = 1$ (the upper limit of F is 1).

(d) $\lim_{x \downarrow x_0} F(x) = F(x_0)$ (F is right continuous).

b) Let x_n be a decreasing sequence.

Let events $C_n = \{X \leq x_n\}$

$$\Rightarrow F(x_n) = P(C_n)$$

From $x_1 > x_2 > \dots$ follows $C_1 \supset C_2 \supset \dots$

$$\lim_{n \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n = \bigcap_{n=1}^{\infty} \{X \leq x_n\}$$

From $n \rightarrow \infty$ follows for any $r \in \mathbb{R}$

$$r \not\leq \lim_{n \rightarrow \infty} x_n.$$

$$\text{Thus: } \lim_{n \rightarrow \infty} P(C_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right) = P(\emptyset) = 0$$

CDF-Def.

$$\Rightarrow \lim_{n \rightarrow \infty} P(C_n) = \lim_{n \rightarrow -\infty} F(x_n) = 0$$

$$\Rightarrow \lim_{x \rightarrow -\infty} F(x) = 0 \quad \square //$$

$$c) \lim_{x \rightarrow \infty} F(x) = 1$$

Let x_n be an increasing sequence. ($\lim_{n \rightarrow \infty} x_n = \infty$)

$$\text{Let } C_n = \{X \leq x_n\}. \quad F(x_n) = P(C_n).$$

$$x_1 < x_2 < \dots \Rightarrow C_1 \subset C_2 \subset \dots$$

$$\lim_{n \rightarrow \infty} C_n = \bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} \{X \leq x_n\}$$



From $n \rightarrow \infty$ follows, that $\lim_{n \rightarrow \infty} \{X \leq x_n\} = \Omega$

$$\rightarrow \lim_{n \rightarrow \infty} P(C_n) = P\left(\bigcup_{n=1}^{\infty} C_n\right) = P(\Omega) = 1$$

And thus:

$$\lim_{n \rightarrow \infty} F(x_n) = 1$$

$$\Rightarrow \lim_{x \rightarrow \infty} F(x) = 1 \quad \square //$$

1.5.11. Let X be a random variable with space $\mathcal{D}$. For $D \subset \mathcal{D}$, recall that the probability induced by X is $P_X(D) = P[\{c : X(c) \in D\}]$. Show that $P_X(D)$ is a probability by showing the following:

- (a) $P_X(\mathcal{D}) = 1$.
- (b) $P_X(D) \geq 0$.
- (c) For a sequence of sets $\{D_n\}$ in $\mathcal{D}$, show that

$$\{c : X(c) \in \bigcup_n D_n\} = \bigcup_n \{c : X(c) \in D_n\}.$$

- (d) Use part (c) to show that if $\{D_n\}$ is sequence of mutually exclusive events, then

$$P_X\left(\bigcup_{n=1}^{\infty} D_n\right) = \sum_{n=1}^{\infty} P_X(D_n).$$

Remark 1.5.2. In a probability theory course, we would show that the σ -field (collection of events) for $\mathcal{D}$ is the smallest σ -field which contains all the open intervals of real numbers; see Exercise 1.3.24. Such a collection of events is sufficiently rich for discrete and continuous random variables. ■

a) $P_X(\mathcal{D}) = 1$, Let Ω be a sample space:

$$\rightarrow P_X(\mathcal{D}) = P[\{\omega \mid X(\omega) \in \mathcal{D}\}]$$

$$\bullet \forall \omega : X(\omega) \in \Omega \Rightarrow X(\omega) \in \mathcal{D}$$

$$\Rightarrow \Omega = \mathcal{D}$$

$$\Rightarrow P_X(\mathcal{D}) = P(\Omega) = 1 \quad \square //$$

sample space

b) Let $A_D = \{\omega \mid X(\omega) \in D\}$, $A_D \in \mathcal{F}$

$$\Rightarrow P(A_D) \geq 0$$

$$\Rightarrow P_X(D) \geq 0 \quad \square //$$

$$c) \{\omega \mid X(\omega) \in \bigcup_n D_n\} = \bigcup_n \{\omega \mid X(\omega) \in D_n\}$$

$$\rightarrow 1. \dots \subseteq \dots$$

$$\rightarrow 2. \dots \supseteq \dots$$

① Let $c_i \in \{\omega \mid X(\omega) \in \bigcup_n D_n\}$

$$\Rightarrow \exists^{21} k : X(c) \in D_k$$

$$\Rightarrow c_i \in \{\omega \mid X(\omega) \in D_k\}$$

Since c_i is in one of these sets,

$$\Rightarrow c_i \in \bigcup_n \{c \mid X(c) \in D_n\} \quad \subseteq \checkmark$$

$$\textcircled{2} \text{ Let } c \in \bigcup_n \{c \mid X(c) \in D_n\}$$

$$\Rightarrow \exists^{>0} k: c_i \in \{c \mid X(c) \in D_k\}$$

$$\Rightarrow X(c) \in D_k$$

Since $X(c)$ is in one of those, it has to also be in the union. $X(c) \in \bigcup_n D_n \quad \supseteq \checkmark$

$\subseteq, \supseteq$

$\rightarrow \square_{//}$

d) $\{D_n\}$, with $D_i \cap D_j = \emptyset, i \neq j$

Suppose $c \in A_i \cap A_j, i \neq j$

$$\Rightarrow c \in A_i (X(c) \in D_i) \wedge c \in A_j (X(c) \in D_j)$$

$$\Rightarrow X(c) \in D_i \cap D_j$$

$\hookrightarrow$ since $D_i \cap D_j = \emptyset$

$$\Rightarrow A_i \cap A_j = \emptyset$$

$$P_X\left(\bigcup_{n=1}^{\infty} D_n\right) = P\left(\{c \mid X(c) \in \bigcup_{n=1}^{\infty} D_n\}\right) \quad \text{Def. of } P_X$$

$$A_n = \{c \mid X(c) \in D_n\}$$

$$= P\left(\bigcup_{n=1}^{\infty} A_n\right)$$

since A_n
mutually &
exclusive

$$= \sum_{n=1}^{\infty} P(A_n)$$

$$= \sum_{n=1}^{\infty} P(\{c \mid X(c) \in D_n\})$$

$$= \sum_{n=1}^{\infty} P_X(D_n)$$

$$\Rightarrow P_X\left(\bigcup_{n=1}^{\infty} D_n\right) = \sum_{n=1}^{\infty} P_X(D_n) \quad \square //$$